

Finly Project Documentation

1. Product overview

Finly is a role-based finance operations application.

- **User side** focuses on day-to-day money management.
- **Admin side** focuses on oversight, governance, and operational controls.

Core capabilities:

- transaction tracking
- category-based budgeting
- trend and category analytics
- logs and audit visibility
- CSV/PDF report exports
- profile and password management

2. Architecture summary

Backend

- Framework: **FastAPI**
- API base: **/api/v1**
- Middleware: request id, cache-control for API responses, CORS, rate limiting
- Routers:
 - **auth_router**
 - **user_router**
 - **category_router**
 - **transaction_router**
 - **budget_router**
 - **analytics_router**
 - **admin_router**
 - **log_router**

Frontend

- Static HTML + modular JS/CSS
- Served by FastAPI from **Frontend/**
- Workspace shell with hash navigation
- Theme toggle (light/dark) with persisted preference

3. Frontend page map

Public/Auth pages

- **index.html** - landing page
- **support.html** - support and FAQ

- `login.html` - sign in
- `signup.html` - create account
- `forgot-password.html` - reset flow step 1
- `reset-password.html` - reset flow step 2

User workspace (`app.html`)

- `#dashboard`
- `#transactions`
- `#reports`
- `#budgets`
- `#notifications`
- `#profile`

Admin workspace (`admin.html`)

- `#dashboard`
- `#users`
- `#transactions`
- `#categories`
- `#reports`
- `#logs`
- `#profile`

4. Auth and recovery flow

Sign in

- user submits email/password
- backend returns token + role context
- frontend routes user to proper workspace

Forgot password (2-step)

1. `forgot-password.html` sends verification code request
2. `reset-password.html` verifies code and sets new password

5. Reporting and exports

Live reports in UI

- KPI metrics: net, inflow, outflow, ratio
- trend chart (net/inflow/outflow modes)
- category distribution chart
- ranked category summary
- monthly snapshot table

Downloadable exports

- **CSV**: tabular metric export
- **PDF**: multi-page executive report that includes:
 - executive snapshot
 - live report summary
 - key signals
 - operational detail tables
 - trend page with notes
 - spending mix page with actionable summary

6. Styling and UX principles

- compact, desktop-first layout
- responsive breakpoints for tablet/mobile
- popup system with structured headers and scrollable bodies
- consistent button hierarchy and spacing
- inline form validation for key entry forms

7. Local development

Prerequisites

- Python 3.x
- Docker + Docker Compose

Start app

```
make dev
```

Other commands

```
make init-db  
make share
```

8. Environment configuration

Use `.env.example` as baseline. Typical keys include:

- database connection variables
- JWT secret
- CORS origins
- rate limit / password reset storage paths
- SMTP settings for password recovery emails

9. Troubleshooting

- If UI changes do not appear, hard refresh (Cmd + Shift + R).

- If report export looks outdated, refresh and re-export to ensure latest JS is loaded.
- If login/reset fails in development, verify API base and environment values.

10. File references

- Backend entrypoint: `app/main.py`
- Workspace shell: `Frontend/scripts/workspace.js`
- User pages: `Frontend/scripts/pages/user.js`
- Admin pages: `Frontend/scripts/pages/admin.js`
- Export engine: `Frontend/scripts/core/export.js`
- Theme/auth helpers: `Frontend/scripts/core/theme.js`, `Frontend/scripts/core/auth-page.js`